H48W35 - Software Development: Graded Unit 2

Speedo Models Web System Evaluation Report

Stage 3

Evaluation.

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Class: HND Software Development 2A

Word Count: 7474

Submission Date: **12-06-2024**

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Introduction

Purpose of this report

The purpose of this evaluation report is to analyse the development and implementation of the Speedo Models multiplatform ecommerce-web project. This report will provide an in-depth evaluation of each stage of the project starting from the initial planning and analysis stage through to design, implementation, and the testing stage. The evaluation will take into consideration the user requirements met, the efficiency of the development process and the overall quality and usability of the final system.
This report will cover the strengths and weaknesses of the development process and the product, including how well the project adheres to the project specifications, if the implementation features work efficiently and if the project plan was followed effectively. This report will also cover the challenges encountered during the project and what strategies and techniques were used to tackle them.

Finally, the evaluation will include recommendations for any further future implementations/additions and any skills and knowledge gained during the life cycle of the project, to understand how the learning outcomes were harnessed to achieve professional and or educational growth as a software engineer.

Terms of Reference

This section will consist of a summary of the scope and objectives will be included in this evaluation. This is to ensure evaluation clarity for all the stakeholders and other users reading this document.

Background and context

Speedo Models is a family run business who sell car collectible models. They are currently operating using an out-of-date paper-based order and purchasing system. They have been looking to computerise this system for a while and have finally taken a step forward to achieve this by hiring a software engineer. The goal is to streamline operations, reduce human error and to provide a platform for both customers and employees to use. Utilising the waterfall software methodology, this project covers all the main stages for a typical

software development project, namely: planning, analysis, design as part of stage one, implementation, testing and external documentation as part of stage two and the evaluation as part of the third and final stage.

Purpose of the evaluation

The main objective of this evaluation is to analyse and conclude the effectiveness and success of the Speedo models website system project. In this evaluation report, we will do the following:

- Determine whether or not the project abided by the requirements specifications and built accordingly with the design documentation as well as any modifications made.
- Identify strengths and weaknesses in each stage of the software development journey, including the final web system product.
- Document skills and knowledge gained throughout the process.
- Give recommendations for future improvements.

Scope of the evaluation

The evaluation will cover the following aspects of the project:

- Functional Requirements
- Non-Functional Requirements
- Constraints
- Design, implementation and testing
- Project management

Evaluation Criteria

The following is the evaluation criteria for the project:

- Functionality - how well the implementation matches the requirements
- Usability - this covers the ease of use of the system, the user interface design and the overall usability
- Performance - the speed and efficiency of the system
- Security - data protection implementation done at the code level, user authorisation, authentication and compliance with data protection laws.

- Project Management - how effective the project plan is and how well I followed the project schedule, managed tasks and handled unforeseen events.

Project Overview

Background

Previously mentioned, Speedo Models sell car models, but to expand, it is a family run business that sell Scalextric racing car models, established in 2002, they were able to grow to one of the top retails in the business.

Project objectives

The main objective of this project was to develop a multiplatform ecommerce web-application for Speedo Models, for a more modern approach to their current ordering and purchasing process. This approach aims to improve the effectiveness of operations, reduce human error and to provide an online platform where both the customer and employees will use. To achieve this, I have focused on developing key goals that encompass the functional and non-functional requirements:

Automating the order process by tracking and fulfilling orders, customer management by allowing specific employee members to edit or delete customer users, product management allows specific employees to edit products and all to edit, delete and add products, user employees have different roles and permissions based on their roles, the different roles include, admin, sales assistant, warehouse assistant, invoice clerk, stock control manager, sales manager and assistant manager. User customers have different membership statuses and depending on the status the customers shipping costs will vary, standard member has to pay £4.99 for shipping, premium member has to pay £2.99 and corporate has to pay £0.99 for shipping. Some more key goals include integrating Stripe as a third-party payment gateway to facilitate online transactions, allowing specific employees to generate daily, weekly and monthly reports for orders, invoices, products and users, allowing specific employees to create and manage invoices and orders.

Project Milestones

Utilising the waterfall software development methodology, the project was broken up into three distinct stages:

The Planning stage, where the initial planning consisted of conducting and documenting interview questions to get a more informed understanding of the customers' needs and

wants followed by a planning report which allowed for me to conduct research on the technology stack, I was planning on using and to create some initial design documents such as an initial use case and initial class diagram. During the initial planning sub section, I also created a project plan with realistic timescales and identifying/organising main milestones, tasks, scheduling the tasks, the deliverables and resources used. After the initial planning, I moved on to creating analysis and design documents, where I created UML diagrams - Use case, class, activity, sequence, entity relationship diagrams, Event handling forms, module descriptions and screen layout charts.

Project Deliverables

As it relates to the requirements specification, there were a list of must haves that required implementation near the top and the lower priority near the bottom. Following this document in order of highest priority I implemented functionalities accordingly. Here is a list of the requirements (*in order of priority from highest at the top to lowest at the bottom*) and their implementation status:

Requirements	Implementation Status
The system must allow company staff to record a new product.	Implemented
The system must allow company staff to generate daily, weekly, monthly and yearly reports.	Implemented
The system must allow company staff to view details of orders placed by a customer.	Implemented
The system must allow company staff to amend/cancel & refund a customer's order.	Mostly Implemented - amend and cancel is implemented
The system must allow customers to place orders.	Implemented

The system must allow customers to make a payment.	Implemented
The system must allow customers to view existing orders.	Implemented
The system must allow customers to amend/cancel orders.	Implemented
The system must allow customers to track an order.	Implemented
The system must allow customers to print return slip.	Implemented
The system will allow all users (registered and unregistered) to view products.	Implemented
The system will allow registered users to select items for purchase.	Implemented
The system will allow all unregistered users to register (new customer)	Implemented
The system will allow all registered users to login (existing customer)	Implemented
A description of the product will be recorded in the product details section.	Implemented
The price of the product will be recorded in the product details section.	Implemented
The stock level of the product will be recorded in the product details section.	Implemented
The system will hold the customers id, name, address, telephone number and order history.	Implemented - although the Id can only be viewed in the Users Report
The system will have the staff user role Admin, who will be able to register new staff and provide Username, first name, last name, email, address, password,	Implemented

company name for domain name (username).	
The system will have the staff user role Sales Assistant who is responsible for processing customer orders, entering of orders, taking payment and customer registration.	**Mostly Implemented - they do not deal with payment.**
The system will have the staff user role Invoice Clerk who is responsible for sending out invoices to corporate customer.	**Implemented**
The system will have the staff user role Sales Manager, who has access to all the data (apart from the ability to register staff)	**Implemented**
The system will have the staff user role Assistant Manager – Same capabilities as a Sales Assistant and can also amend the price of products and have control over stock.	**Implemented**
The system will have the staff user role Stock Control Manager – who is responsible for stock control.	**Implemented**
The system will have the staff user role Warehouse Assistants who are responsible for the dispatch of goods to customers.	**Implemented**
All users require a user ID and password.	**Implemented**
The system will alert customers and staff when the stock of an item is low.	**Implemented partially - only alerts customers**

The system will have a section in the contact us page where they can query about stock availability.	Not Implemented
The system implements payment gates: stripe	Implemented
The system will have a loyalty program for discounts, based on the purchase history of the customer.	Not Implemented

I have implemented the majority of the requirements, but there was still room to implement more, I also, have other implementations that have not been recorded because they were not mentioned as a must have in the requirements specifications, but I thought they were essential to add: such as adding staff salaries, deleting users, allowing the admin staff to have access to every functionality. However next time I will pay close attention and focus on executing the implementation of the given requirements.

Initial Planning

The initial planning stage consisted of a production of a report that included, the aims of the project assignment, information gathering such as interview questions and background research to clarify the brief, my interpretation of the assignment brief, initial functional requirements, non-functional requirements, constraints, initial top-level view of use case models, a class diagram, identifying resources, materials required and identifying information sources to be used. After this analysis of the project assignment brief, I moved on to creating a project plan that identified: main milestones, a breakdown of tasks at each stage, a realistic timescale for each task, scheduled tasks, documented deliverables and resources used.
During this stage and following the discussed guidelines, I managed to produce a strong report that I would later reference during the later stages of the development process.
Throughout the planning process there are both strengths and weaknesses:
Strengths

I created and followed, clear and concise project aims, which would act as the foundation for the project. I conducted thorough background research on competitors, technology stacks and by asking over 40 client interview questions utilising a variety of interviewing techniques, ensuring open-ended questions to get the most out of the interview. I created a detailed initial requirement specification detailing both functional and non-functional requirements off the back of the project brief and the interview questions. As for the planning schedule, I clearly identified key milestones such as Implementation, testing and deployment, I had a detailed breakdown of the tasks, a realistic timescale to ensure a feasible project timeline, good resource allocation to optimize resource use for each scheduled task, I had clearly defined project deliverables at the end of each stage and identified resources such as technology tools for development and for internal and external documentation.

Weakness

As a result of time and resource constraints the project does lack some features that would have made it better. There was limited user feedback as a result of following the waterfall approach. In the initial high level use case I forgot to make it so that the admin has full access to all functionalities. In the planning schedule, I overlooked certain resources required which I later had to implement, and the task scheduling was somewhat rigid, which made it harder to adapt to sudden changes and vulnerable to delays.

Modifications

In the initial use case model, I had more details then required, so before submitting I recreated it so that it lacked detail, as it is an initial use case model and not a detailed one.

After reviewing the project schedule, I had realised I had missed out a resource “Canva” – which was meant to be used for the main concepts of the front-end design, so I implemented it into the resources section, before submitting the project plan.

Knowledge and Skills

Throughout this process alone I gained analytical skills as it relates to translating project briefs into an actionable project plan. Diving deep into collecting more information to strengthen the project brief I learned an array of fact-finding techniques; such include observation and the problems that comes with it such as the Hawthorne Effect. I improved my time management skills throughout this process as I had to estimate the

realistic amount of work I could complete in the given time, and I also learned to adapt the project plan and schedule to accommodate changes to some of the end dates that took place near the middle of the project.

Conclusion

To conclude the initial planning stage acted as a foundation for which to build the application and a good document to refer back to if I ever forget about specifications required. I gained a good understanding of which technology best suited me through decent research, this allowed me to self-analyse my current skillset and choose the technology stack that best suited my current knowledge base and perhaps use the different technology stacks I found during my research for future projects.

Analysis & Design

This stage consisted of creating diagrammed documents and compiling them together into a single word document, this included the following for placing an order and logging in functions: detailed Use case diagrams, activity diagrams, sequence diagrams and event handling forms. I normalized an order form and used that to create an entity relationship diagram, I created a detailed class diagram that covered all the classes to be created (expanded on the classes during implementation), I created a Module Description for a calculate total method I made and the default dispose method which is part of the IDisposable Interface in C#. I then prototyped the front end in ASP.NET C# using html, CSS and bootstrap so that I could take screenshots and create the screen layouts chart for each page of the web application.

Strengths

All the diagrams: use case, class, activity and sequence provide a clear and concise breakdown of the systems functionalities which set up the implementation stage very well. A decent normalised form coupled with a good class diagram structure, since I was using entity framework code first approach for database construction, it helped reduce repeating and redundant data particularly when it came to creating the order class. The Event handling forms allowed me to get a holistic view of how the methods were to be named and what main functionalities will need to be incorporated into the system. The user-centric screen layouts were prototyped effectively and allowed me to gather and see all the pages that were going to be used in the application.

Weakness

The class diagram in particular, although set me up for a solid implementation stage, still lacked a class or two (such as the Parcel class) that were later implemented, again due to building the solution using the waterfall software methodology I was restricted to go back and add missing classes to the diagram which would of given me an easier time when it came to visualising the missing classes relationships with the other classes in the project, however, this is nothing that wasn't fixed and overcome in the implementation stage.

Modifications

For the placing order sequence diagram, I realised it didn't make sense for a web application; I had created another one with more detail including a checkout, discount, payment and Order confirmation page that had previously been missing and instead the original design had a standalone desktop format with order form, order manager and an order page instead. I managed to make this change after I had implemented the screen layouts because I had prototyped the front-end and therefore had a better Idea as to how the user would place an order.

Also near the end of the analysis and design stage, decided to use https://uk.scalextric.com/ as an inspiration for specific elements as it relates to the front end – such as the way the categories are laid out in the home page.

Knowledge and Skills

Although I already had skills in utilising diagramming techniques with UML diagrams, it helped sharpen this skill set and I also gained a fond new respect for the importance of the UML diagrams in relation to it assisting the overall success of the application development process. With database normalisation and Entity relationship building I developed a deeper understanding of both processes, I even looked into going past the third normalised form and read about the fourth normalised form, although I never implemented it, just educated myself on it. I enhanced my skills in creating user-friendly interfaces by utilising key technologies such as bootstraps pre-built templates to rapidly develop the front end.

Conclusion

To conclude overall the analysis and diagram stage set up the implementation stage for success, with a lot of documentation to refer to when building the back end and dynamic front-end.

Implementation

At the implementation stage I used the visual studio community 2022 IDE, C# for the programming language and the ASP.NET MVC5 framework, with Entity framework and identity security for user authentication and authorisation. Using the documentation from the previous stage and common sense I implemented the business model code listings for core functionalities including employee, customer, product, invoice and order managers in the controller with equivalent business POCO (Plain Old Clear Objects) classes in the model's folder. I also implemented the View Model Code Listings, included developing the user interface dynamically for pages such as employee registration, recording products, the payment page amongst many other pages. For unfamiliar libraries I integrated third-party API's such as Stripe to handle payment processing, Twilio for both SMS and Email order, account confirmation set-up and SMS for enabling user authentication, I used the random class library to generate random products for the feature page, I used GitHub to push/commit changes as another source to store the code, I took advantage of different online bootstrap templates using https://mdbootstrap.com, I also created a new way to deal with redirecting unauthorized users when they try to enter an invalid routing URL like "/Admin/Index" for example, this will redirect them to the home page instead of the login page if they are already logged in, which was the issue I found with the already built in "Authorize" data filter. The custom attributes (found in the 'CustomAttributes' folder in the project visual studio solution) are split into two, there is one to redirect unauthorised users to a specific URL and another which redirects unauthorised staff users depending on their roles.

I implemented robust error handling on methods where necessary, using try catches for exceptions and if statements to find empty (null) objects and return http bad requests to the action result, I also used validation and anti-forgery token checks in both the front-end (the view) using HTML helpers with razor syntax and in the back-end (controllers) using data/action filters. I also implemented internal documentation using regular comments and XML documentation throughout the full MVC project, The XML documentation is automatically exported to its own document when you build the solution, I also ensured consistent naming conventions and appropriate use of indentation.

I planned to implement PayPal API for payment processing and Gmail for email registration and login integration, but due to time constraints, never got around to doing so.

Along the way I used tools such as stack overflow and ChatGPT 4 to help understand any problems I ran into, so that I can create or come up with a solution as it relates to my problem domain.

Strengths

The business model code I implemented effectively handled the main functionalities for the application and was split up appropriately to follow the MVC format, which ensured data integrity. The implementation of regular methods to deal with repetitive functions also helped throughout the process. Thanks to utilising bootstrap templates the views provided a responsive and user-friendly user interface and therefore enhancing the user experience. I ensured effective error handling and managed exceptions a validated user inputs throughout the whole project. I had clear and concise documentation, with detailed comments and consistent formatting, to ensure good readability and maintainability.

Weakness

I did not implement custom error pages, which can reduce the overall user experience. As mentioned previously, I also did not fully implement all the third-party integrations I had planned. Some areas of the code could benefit from further optimization for performance improvements. Some of the views HTML code does not have the Razor syntax when dealing with dynamic implementation and therefore can look out of place or messy when comparing all the views.

Modifications

As it relates to the design stage, I made changes to relationships between classes and even added a class that did not exist in the class diagram – the Parcel class was added during implementation. After implementing certain parts of the code, I went back and integrated try and catches, which is why some methods have an if statement whilst others have try and catches. I went back through the code and modified database queries with LINQ methods such as the use of “Select” instead of running the objects through for loops, this was to improve readability of the code. At the end of the implementation, I read through the code and extended the internal documentation by just adding more comments to statements in methods where I had forgotten to comment.

Knowledge and Skills

I gained more in-depth knowledge about ASP.NET MVC Framework and how to divide the logic for optimal performance.

I continued to improve my skills in Entity Framework, using the code first approach especially as it relates querying the database and knowing when to use asynchronous methods and when not to.

I gained new knowledge when it comes to integrating third-party API's in a coding environment.

I also gained new knowledge and skills when it came to integrating bootstrap templates to create user-friendly UI's.

I learned how to integrate unfamiliar libraries effectively to fully optimize the system functionality.

Conclusion

To conclude this section of the project provided me with the biggest learning curves and confidence when it comes to problem solving, software development and programming as a whole. I learnt the importance of not getting too carried away when working on a software project in a pressured environment and under time constraints.

Testing

The testing stage ensures that the system is fully functional with no - to minimum bugs in the final deployed application. With testing I created a Test Strategy that covers what strategy I used, which was both black and white box testing. For the black box testing I used the boundary value analysis technique, which was embedded in the completed list of test data (Test Logs), and for white box testing I used unit testing which Is found in the test strategy documentation. I also created a list of test cases, which documented the number of test cases and the purpose of each test data. The completed Test Data documented (this is the Test Logs word document) included expected and actual results with a print screen layout in each "actual result", with boundary value analysis being recorded depending on the test case. I created Error Logs for any errors I ran into which also have screenshots of both the error and the fix – I managed to fix all the errors I had ran into. I created a system document which recorded the test number, the date followed by a pass or fail, with a reason for failure if any failed – since non failed I didn't

have to fill this field in. The system documentation is to be used alongside the Test logs and list of Test Cases when referring to the test data, as all three of these documents share the same test data.

Strengths

The test strategies used is well known in the industry and in relation to boundary testing, it allowed me to test all aspects of the web application, to ensure that the application was bug free.

Detailed test cases allowed me to document and understand the purpose of each test case and again ensuring a comprehensive coverage of testing the applications functions.

The Test logs where expected and actual results can be viewed visually provided clear evidence of the test outcomes.

Effectively creating error logs with both visually and documenting the data provides a detailed and useful way to track any issues that had come up and how they were fixed.

Weakness

Although I did implement unit testing, I only tested two methods and documented it manually, I could have automated the process during the development stage as visual studio ASP.NET has an inbuilt testing suite that I could have taken advantage of and ran more tests.

Modifications

The Test Strategy document was just purely going to be a word doc where I explained the testing strategies used in the different testing documents. It ended up evolving to include the white-box unit testing technique, which was implemented after the test strategy had been created.

Knowledge and Skills

I developed my skills further when it comes to creating Test Cases and sharpened my ability to spot potential test cases and effectively document them in a way I, or anyone reading it can understand.

Manually writing out the test cases, data and system documentation improved my skills in performing manual testing.

Error logging improved my ability to spot and fix errors and communicate how I did so in the documentation.

Iterating through test cases allowed me to find issues that I didn’t find upon my first test run through and this lead to more bug fixes, especially with logic errors.

Conclusion

To conclude the testing section, Testing is where I sharpened my problem solving even further, spotting and fixing errors in the code whether that was because I was calling and displaying the wrong type of object in the view or fixing the logic of a method, I found testing very useful and helpful in debugging the application.

External Documentation

During the creation of the external documentation, I had to create a detailed User Manual, which was developed as a windows form application using the .NET framework and used VLC media player to run tutorial videos. The application was split into two, one customer user manual, the other was an employee user manual, both covered all angles as it relates to the main user functionalities such as placing an order, logging in, registering, viewing their own orders for customers or recording a products as an employee, each employee role has a video tutorial related to their given role. The external documentation also included a robust installation guide, which was a step-by-step guide for system requirements depending on device being used, including Microsoft edge, google chrome and safari (for IOS devices), there was also an extra information and support section. The updated Gantt Chart recorded the changes to the timeline after it had been extended and the extra resources used, which in this case was the addition of visual studio windows form.

Strengths

The user manual was thorough, covered all user functionality, provides clear step by step instructions and to optimise performance I used VLC for up to date and fast video playbacks and the application recycles videos if there are repetitive videos per staff role for the employee user manual – These features provide visual aid for the user and in turn will allow them to get use to the system and navigate the web application with ease in no time.

The installation guide provides a detailed and easy to follow bullet points list, which ensures the user knows how to get access to the web application.

The updated Gantt chart gives a more accurate reflection of the current project's progress, which improves the overall project plan.

Weakness

The user manual lacks a trouble-shooting guide, which limits the users to fix any major/ complicated issue independently.
If the user manual is not updated to run on modern systems, then it might not work, which means going back and forth with the software engineer (me) to keep the standalone application updated.
I spent a lot of time on the development of the application and that time could have potentially been best used for the evaluation report.

Modifications

The user manual was originally just going to be a detailed technical written manual but later evolved to a video format.
After the video format was implemented I originally used windows media player, which allowed the user to pause, play, rewind and go forward as party of the media player functionality, but this system was outdate in visual studio and therefore had complications and issues running smoothly and so I migrated to using VLC to run the videos, but as a result lost the capabilities to play, pause, rewind and go forward, so I had to implement the pause, play and video ending functionalities programmatically.

Knowledge and Skills

I further improved my coding skills utilising concepts like interfaces and splitting up solutions under the same project solution on visual studio community 2022 and integrating video media player into an application.
I gained experience writing technical documentation for the installation guide, which gave me insight into thinking from the end-user's perspective and ensuring the instructions were easy to follow.
I also developed my skills using OBS recording service to configure and record the video tutorials appropriately.

Conclusion

To conclude the External Documentation process is important for the proper use of the Speedo Models web system, for both the customer and for all the employees, covering information required for a successful and enhanced user-friendly experience.

Other Criteria for Evaluation

User Requirements

The functional user requirements have already been mentioned in the project deliverables. The user requirements are the functional and non-functional requirements of the system. Here is a better visual break down of all the requirements:

Functional Requirements:

1. The system must allow all company staff to record a new product.
2. The system must allow all company staff to generate daily, weekly, monthly and yearly reports.
3. The system must allow all company staff to view details of orders placed by a customer.
4. The system must allow all company staff to amend/cancel & refund a customer order.
5. The system must allow customer to place orders.
6. The system must allow customer to make a payment.
7. The system must allow customer to view existing orders.
8. The system must allow customer to amend/cancel orders.
9. The system must allow customer to track an order.
10. The system must allow customer to print returns slip.
11. The system must allow all users to view products.
12. The system must allow all users to select items for purchase.
13. The system must allow all users to register as a customer.
14. The system must allow all users to login as a customer.
15. A description of the product will be recorded.
16. The price of the product will be recorded.
17. The stock level of the product will be recorded.
18. The system will hold the customer's number/id.
19. The system will hold the customer's name.
20. The system will hold the customer's address.
21. The system will hold the customer's telephone number.
22. The system will hold the customer's order history.
23. The system will have staff role Admin, who will register a new staff.
24. The system will have staff role Sales Assistant, who will be responsible for processing customers orders, entering of orders, taking payment and customer registration.
25. The system will have staff role Invoice Clerk, who will be responsible for sending out invoices to corporate customers.
26. The system will have staff role Sales Manager who will have access to all data.
27. The system will have staff role Assistant Manager who has the same access as a sales assistant and can amend the price and adjust the stock level of products.
28. The system will have staff role Stock Control Manager, who is responsible for stock control.
29. The system will have staff role Warehouse Assistant, who is responsible for the dispatch of goods to customers.

30. The system will allow admins to register new members of staff and provide their username.
31. The system will allow admins to register new members of staff and provide their first name.
32. The system will allow admins to register new members of staff and provide their last name.
33. The system will allow admins to register new members of staff and provide their email.
34. The system will allow admins to register new members of staff and provide their address.
35. The system will allow admins to register new members of staff and provide their password.
36. The system will allow admins to register new members of staff and provide their company name for the domain name.
37. All users require a user ID and password.
38. The system will alert customers & staff when the stock of an item is low.
39. The system will have a section in the contact us page where they can query about stock availability.
40. The system implements stripe payments.
41. The system will have a loyalty program for discounts, based on the purchase history of the customer.

Out of the 41 requirements, 39 were implemented with 3 of the 39 being partially/mostly implemented. As a result, the system functions very closely to what the client had envisioned for the system.

Here is a list of the Non-Functional Requirements:

1. The system will have information about the computing equipment stocked by the company.
2. The system needs to comply with relevant legislations such as data protection laws (GDPR) and online trading regulations.
3. The system must maintain compatibility with widespread legacy and contemporary desktop browsers.
4. The systems design must include a user-friendly interface, responsive layout, and visually appealing graphics consistent with the company's branding.
5. The system will include company logos, images and branding elements as part of the brand recognition.
6. The system will be compatible with various devices: desktops, laptops, tablets and smartphones to cater to diverse user preferences.
7. The system will record sales for inventory management, financial reporting and customer service purposes.
8. The system will have security measure such as data encryption, secure payment gateways, and user authentication protocols will be implemented to protect sensitive information.

Ease of Maintenance

As the web application was built in an Object-Oriented Programming language and utilised best programming practices, the system has a strong foundation and structure, as well as extensive documentation to refer back to and therefore any updates, or changes required in the future will be very straightforward to implement.

Recommendations

Although most of the requirements were met, there were still a variety of recommendations that could be made for the implementation of the Speedo Models web application and to the process of the development.

One recommendation for a future implementation, would be the integration of further third-party services such as Gmail and Facebook for both account creation and logging in as well as further integrations as it relates to payment gateways like PayPal and shipping providers. The reason for this is to enhance the overall user experience for both the staff and customers by streamlining business processes and improving how data communicates across the web.

Another recommendation for the future of the web application which has been mentioned previously, and that's to implement a custom error page, as this improves user experience by offering concise and helpful messages for the user to understand clearly.

The incorporation of AI and Machine Learning as a future recommendation could offer a variety of different utility to enhance the user experience, such as offering personalized product recommendations and automating customer services with chatbots that offer quick solutions, this integration provides an efficient experience for the customer.

Another recommendation for future integration, that would work well with personalized product recommendations offered with the AI and Machine Learning incorporation, would be the creation of a product rating system, that allows customers to leave a review and a comment on the product they have purchased, this will not only further enhance the user experience for the current customer, but also for future customers and can help convert a viewer to a customer whom is browsing and is unsure about the company as it enhances the companies credibility.

Integrating a comprehensive loyalty program that includes rewards, discount and exclusive offers for customers based on their membership status and how often they purchase from the company – if well designed this will increase customer retention, encourage future purchases and build long-term trust between the customer and the company.

A final recommendation would be to hire a team of software engineers/developers rather than just one, this also depends on budget, but if it is within the budget, this will speed up the development process and allow the company to adapt their computerised system quickly and efficiently.

Project Diary

Green = Positive comments.
Other Colours = General comments.

Date	Topic	Comments
07/02/2024	1.1.1.2 – Interview Questions	Uploaded interview questions.
15/02/2024	1.1.1.2 – Interview Questions	Heard back from interview questions. The way question 12 was answered confused me so I clarified it with Dana.
22/02/24	1.1.2. - Project Plan	Struggled with showing critical paths in 'Microsoft Project'
29/02/24	1.1.1.4 - Requirements Specification	Looked into but didn't implement MoSCoW prioritisation technique.
07/03/24	1.2 - Analysis and Design	Created two sequence diagrams, as I was unfamiliar with the way to create it - so I did

		some more research on the topic and came to a better conclusion on which one to use
12/03/34	1.2 - Analysis and Design	Struggled with implementing the dropdown list for “shop” navigational property. Perhaps the code for the functionality is outdated and bootstrap no longer supports the function.
12/03/24	1.2 - Analysis and Design	Didn’t like adding the background text for the front-page image - this is for the screen layouts (might revisit)
18/03/24	1.2 - Analysis and Design	**Activity Diagram** - I changed the place order activity diagram so that it doesn’t include login and just assumes a customer is logged in, as I have a separate login activity diagram.
24/03/24	1.2 - Analysis and Design	**Implemented everything, just on time.**
01/04/24	2.1 - Implementation	Spent 6 hours on the basket functionality. Although I got it to work, I still have to add Stock Level updates to it and

		I'm not finished. Could of watched a video and implemented an easier basket functionality such as storing it in sessions rather than building a full model for it. Also could of watched tutorials to complete this functionality quicker.
02/04/24	2.1 - Implementation	Created another action filter attribute overriding the **Authorize Attribute** class and overriding the **AuthorizeCore** and **HandleUnauthorizedRequest** methods. I called the class: **AuthorizeOrRedirectAttribute**.
05/04/24	2.1. - Implementation	I had an issue learning to implement jQuery for dynamic model update when it came to updating my orderliness in the basket page. So I instead just used forms, which refreshes the page (which I was trying to avoid) – **might revisit this at a later date if I have time**.

06/04/24	2.1. - Implementation	I had issues trying to load the basket page whilst maintaining the scroll position - **revisit this at a later date if I have time**.
12/04/24	2.1. – Implementation - Payments	I was going to implement my own credit card/payment system but decided to go with stripe instead. I had already made the models, classes etc. Also going to try and implement PayPal at a later date.
16/04/24	2.1. – Implementation – Async and Try Catch	Converted the checkout, payment and some of the shop controllers to Async as I just learned that this is important to do with methods dealing with updates/changes to the DB: I/O bound operations. Also, since many unknown errors can occur during runtime, using try catch to 1 deal with error efficiently and 2 ensure user friendly error pages.
22/04/24	2.1. – Implementation – Email Confirmation	I struggled to quickly implement this. I tried speed reading through the documentation, but

		parts of the code were outdated and not working, unlike the SMS implementation which was seamless and easy to add. I will return to this at a later date if there is time. **I ENDED UP IMPLEMENTING IT!**
01/05/24	2.1. – Implementation – Amend order	I struggled to develop this, as I was not sure to either send the order or order line to the POST and so experimented a bit for 3 days before selecting the order model to throw.
07/05/24	2.1. – Implementation – Recycling Pages	Instead of having duplicated layouts and separate pages/ account controllers per user role, I have created one controller and made my own custom action filter, to deal with the authorization correctly per page access. This has allowed me to reduce duplication of HTML pages and saving memory and computer resources.
13/05/24	2.1. – Implementation – Return Slip	I was going to make this a downloadable PDF only,

		but I decided for it to be an un-editable HTML page, but the user can print/download it and fill it out – note to self I made it myself inspired by the “Return Packing Slip” page for Ikea, screenshot found in GU2 folder. I also managed to change the window print function provided to use by JavaScript(or bootstrap) library and made it so that it doesn’t include the navigational properties or buttons.
23/05/24	2.1. – Implementation – XML Documentation	Managed to implement this to the overall project (which means I need to add summary tags throughout every actionresult method and class in the project), and a XML documentation is produced when I “Build” the project/solution.
25/05/24	2.2 – Testing – Test Strategy	When testing the application, we will do both Blackbox and Whitebox testing. For the Blackbox, we will integrate the Boundary

		Value Analysis (BVA) technique, in the 'Completed Test Data, Expected Results and Actual Results' section. For the Whitebox testing we will create Unit Test following the Microsoft tutorial on how to do it in ASP.NET MVC Applications, for three different controllers to View Data and Action Result returned by a controller. Testing Error Logs, I will record whenever I run into a problem.
02/06/24	2.3 – External Documentation – User Manual.	I decided to implement the user manual as a windows form application, where there will be links to various video tutorials dependant on whether a user is a customer or employee.
03/06/24	2.1 - Implementation	I decided to ensure the admin dashboard allows for all functions, this means when I create the user manual, I can create all the employee user function in one area.

04/06/24	2.1/2.2	Completed testing two days before deadline.
10/06/24	3 - Evaluation	Started the evaluation a bit later than I anticipated.
12/06/24	3 - Evaluation	**COMPLETED.**

Conclusion

To conclude when I look back and analyse each stage, I can take lessons and skills gained from each and every one of them. The full process of creating this project has been a very insightful and lesson filled journey. This process has given me newfound confidence coupled with a deep understanding of the software development process.
Through all the stages the most difficult has to be the implementation - building the application, although straight forward, took the most time, however if it was not for the other stages the end-product would not have been as well-rounded and function nearly as well.
The main skills acquired through the process was problem-solving as it relates to building a multiplatform web-application – although a challenge, I overcame all the obstacles from UI flaws to back-end errors, I was able to stay disciplined throughout the process and put a lot of my theoretical knowledge to the test. Going forward I will be utilizing my software development skill set to work on and build more web-based applications.

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